

Venkata Saratchandra Patnaik Markonda Patnaikuni

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EDUCATION

Master of Science in Computer Science (CGPA: 3.81/4)

December 2025

Arizona State University, Tempe, AZ.

TECHNICAL SKILLS

Programming & Scripting Languages: Python, R, Java, Kotlin, Go, JavaScript, TypeScript, C

Web & Mobile Development: React.js, Node.js, Express.js, HTML/CSS, Bootstrap, Android Studio, Flask, REST APIs

Databases: MySQL, PostgreSQL, Google Firebase, MariaDB

Cloud & DevOps Tools: AWS(S3, EKS, EC2, Greengrass, Lambda, Load Balancer), Docker, Kubernetes, Linux, Shell Scripting

CI/CD, Monitoring & Dev Tools: GitHub, GitOps, ArgoCD, Jira, Jenkins, Grafana

Machine Learning & AI Frameworks: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, Matplotlib, OpenAI API

PROFESSIONAL EXPERIENCE

Software Implementation Engineer: Amagi Media Labs, Bengaluru, India

August 2022 - November 2023

- **Solution Architecture & Automation:** Designed onboarding solutions for content creators, **optimized video ingestion and distribution**, and used **Terraform**-based automation for customer onboarding, reducing manual setup time by **90%**.
- **Cloud & Deployment:** Managed 15+ **containerized Microservices** on **AWS EKS** with **Docker & Kubernetes**, increasing deployment frequency by **30%**. Worked extensively in **Linux** environments to deploy, monitor, and troubleshoot systems.
- **Scripting & CI/CD & GitOps:** Deployed **GitOps**-based pipelines via **ArgoCD**, enhancing **CI/CD reliability** and rollback efficiency. Developed **Python** scripts and integrated **REST APIs** for video **preprocessing** and verification before **CDN delivery**, reducing failures by **95%**.

Web Development Intern: Blueplanet Solutions Inc., India

April 2021 - June 2021

- **Optimized MySQL database** using stored procedures for 'Campus Club', improving query speed by **50%** and enabling faster data access for 1,000+ student profiles.
- **Enhanced a responsive student search UI** with HTML, CSS, and JavaScript, reducing search time by **60%** and improving user experience.
- **Analyzed web server logs** to identify memory leaks, enhancing application stability by 30% and reducing server resource usage.

PROJECTS

Real-Time Face Recognition on AWS Greengrass (Cloud Computing)

- Developed an edge-based face recognition system using AWS IoT Greengrass, MQTT, Lambda, and SQS, simulating smart camera-to-core communication over EC2.
- Deployed MTCNN and FaceNet models under pip-free constraints by integrating raw PyTorch source, optimizing runtime within AWS Greengrass component limits.
- Debugged complex distributed pipeline using CLI tools and logs across Greengrass, MQTT, and Lambda; automated deployments via shell scripts and passed 100+ real-time autograder test cases.

Plethora - A Fest Management System (React.js, Node.js, MariaDB)

- Built a full-stack event platform for event discovery, registration, and authentication.
- **Enhanced MariaDB database performance** through in-depth query analysis and index optimization, decreasing real-time **query response time** by **50%**.

Expense Tracker (Kotlin, Java, and Google Firebase)

- Developed a mobile expense tracker application using **Android Studio**; enabled real-time budget monitoring and **Improved Firebase synchronization efficiency** by **33%**, resulting in faster data updates and a smoother user experience.